ZEERBOX · PROJECT PLAN

ZeerBox project plan

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First plan (kit 1.9.1 upgrade) at stage P3 with the reference list of every source cited in the repo

ZeerBox project plan

ZeerBox is at stage P3 (design outputs, TRL 3): the constructable design is modelled at detail level D3, drawn, priced and planned, with the pipeline documents in place. The next gate is P4 (verification, TRL 4), which is on hold while the portfolio is capped at P3; before it, Amish is asked to choose the concept (ZBX-DDR-004, all three options within 10 %) and to decide the no-power vent (ZBX-DEC-001 D3). Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Problem and users	Project lead	PRB with users-and-needs and stakeholder tables; URS	Done on paper (ZBX-PRB-001 v0.8, ZBX-URS-001); not validated with users
STEM check	Design lead	ZBX-STM-001 with `stem_check.py`	Open: six option A rows need supplier or field data, or Amish's decision (D6)
Concept options	Design lead	ZBX-OPT-001, ZBX-DDR-004, models of options A, B and C at D3	Done; choice awaiting Amish
Requirements and traceability	Design lead	ZBX-PRS-001, ZBX-RTM-001	Done; PR-003 at risk
Calculations	Design lead	ZBX-CAL-001 v0.4	Done (load fuse corrected, D1)
Model, drawings, BOM	Design lead	`cad/src/model.py` (D3, `cad/fidelity.json`), ZBX-DWG-001 and build plan sheets, `bom/bom.csv`	Done
Build plan	Design lead	ZBX-BLD-001	Done (TRL 3 paper plan)
Regulatory, feasibility, risk, market	Project lead	ZBX-REG-001, ZBX-FEA-001, ZBX-RSK-001 (with design FMEA), ZBX-MKT-001	First issue; Mali rules to assess
Partner and co-design	Project lead	Approach MIT D-Lab's evaporative cooling team; co-design sessions in a dry-season vegetable area of Mali	Not started (decided as first candidate, ZBX-DEC-001)
Product renders	Coordinator	Cycles renders and storefront images from `cad/src/product_model.py`	Views defined; rendering on Amish's Mac

Resources needed

Table 2. Resources (estimates; no cost committed).

Resource	Deliverable	Effort	Estimated cost (USD)
Partner field coordinator (co-design)	Two co-design sessions with growers and a farmer group; revised URS and PRS	10 days	To assess with the partner
Local mason and carpenter (P4, on hold)	One prototype store from ZBX-BLD-001	About 15 days	Labor to assess; materials about 574 (structure)
Local electrician (P4, on hold)	Kit installation and wiring	2 days	Labor to assess; kit about 300
Supplier data	Fan curve, pad effectiveness and pressure drop, pad price in the region	2 days of enquiries	None
Logging through one dry season (P4 and P5, on hold)	Store and supply temperature and RH, airflow, water and energy against PR-002 to PR-008	One season	Within the kit (the controller logs)

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB with users and stakeholders, STM, OPT, REG, FEA, PLN	Met (2026-10-24)
P1 Definition	URS, PRS draft, RSK with hazard analysis, MKT, REG complete	Met on paper (2026-10-24); REG has rows to assess
P2 Design inputs	PRS with targets, tolerance and verification; CAL; RTM; concept selection DDR; STEM check closed; feasibility closed	Met except the STEM check, which stays open on five data-dependent rows (allowed at TRL 3 for a repo upgraded to kit 1.8, STANDARDS section 20)
P3 Design outputs	Model, STEP, drawings, priced BOM, build plan, DEC, design FMEA, traceability	Met (2026-10-24)
P4 Verification	Test plans and reports, verification report, reliability plan	On hold (portfolio cap P3)
P5 Validation	Field trial with a partner, validation and human factors reports	On hold
P6 Commercialization	Launch plan, instructions for use, training; a named deployment partner	On hold

Next steps and decision requested

1. **Decision for Amish: concept choice** (ZBX-DDR-004). Options A (190), B (173) and C (173 of 235) are all within 10 %, so all three go to Amish without a pick. Option A is the current design and stays unless he decides otherwise.
2. **Decision for Amish: no-power ventilation** (ZBX-DEC-001 D3). Recommended: a meshed passive vent through the ceiling with option C's turbine head, so the store keeps a minimum air change when the battery is flat.
3. Ask the fan and pad suppliers for their curves (ZBX-DEC-001 to confirm 2) and rerun ZBX-CAL-001.

4. Approach the first candidate partner and plan co-design sessions before any TRL 4 work.
5. Ready for TRL 4 on paper once 1 and 2 are decided; TRL 4 itself waits for the portfolio cap to lift.

References

Every source cited in this repo's documents, Chicago style.

- ASHRAE. *ASHRAE Handbook: Fundamentals*. Atlanta: ASHRAE, 2017.
- Cigarruista Solis, Luis, Miguel Chen Austin, Euclides Deago, Guillermo Lopez and Nacari Marin-Calvo. "Rice Husk-Based Insulators: Manufacturing Process and Thermal Potential Assessment." *Materials* 17 (2024). https://pmc.ncbi.nlm.nih.gov/articles/PMC11173955. Accessed October 24, 2026.
- ColdHubs. https://coldhubs.com/. Cited in ZBX-PRB-001.
- Efficiency for Access. *Lessons Learned from Implementing Solar-Powered Walk-In Cold Rooms for Smallholder Agriculture in Zimbabwe and Malawi*. https://efficiencyforaccess.org/wp-content/uploads/Lessons-learned-from-implementing-solar-powered-walk-in-cold-rooms-for-smallholder-agriculture-in-Zimbabwe-and-Malawi.pdf. Cited in ZBX-PRB-001.
- Engineering for Change. "Evaporative Cooling Chambers." https://www.engineeringforchange.org/solutions/product/evaporative-cooling-chambers/. Accessed October 24, 2026.
- *e-planet*. "Storage Study of Zero Energy Cool Chamber." *e-planet* 18, no. 2. https://www.e-planet.co.in/images/Publication/vol-18-2/storage.pdf. Cited in ZBX-PRB-001.
- FAO. *The State of Food and Agriculture 2019*. Rome: FAO, 2019. https://www.fao.org/interactive/state-of-food-agriculture/2019/en/.
- FAO and WHO, Codex Alimentarius. *Code of Hygienic Practice for Fresh Fruits and Vegetables*, CXC 53-2003. https://www.fao.org/fao-who-codexalimentarius/codex-texts/codes-of-practice/. Accessed October 24, 2026.
- Fans Online. "Bradford Windmaster, Natural Roof Ventilator, 300 mm." https://fansonline.com.au/product/csr-windmaster-whirlybird-deep-ocean-finish. Accessed October 24, 2026.
- Global Agriculture. "FAO: 14 % of the World's Food Is Lost Between Harvest and Retail." https://www.globalagriculture.org/news/fao-14-of-the-worlds-food-is-lost-between-harvest-and-retail/. Cited in ZBX-PRB-001.
- Howdy Brewer. "CoolBot Walk-In Cooler Controller." https://howdybrewer.com/products/coolbot-cooler-controller. Accessed October 24, 2026.
- IEC 60364-5-52:2009. *Low-Voltage Electrical Installations, Part 5-52: Selection and Erection of Electrical Equipment, Wiring Systems*. Geneva: IEC.
- IFPRI. "ColdHubs: Addressing a Crucial Problem of Food Loss in Nigeria with Solar-Powered Refrigeration." https://www.ifpri.org/blog/coldhubs-addressing-crucial-problem-food-loss-nigeria-solar-powered-refrigeration/. Cited in ZBX-PRB-001.
- IndiaMART. "Cellulose Pad." https://m.indiamart.com/impcat/cellulose-pad.html. Accessed October 24, 2026.
- Kader, Adel A., ed. *Postharvest Technology of Horticultural Crops*. 3rd ed. Oakland: University of California Agriculture and Natural Resources, Publication 3311, 2002.
- MIT D-Lab. "Evaporative Cooling for Vegetable Preservation." https://d-lab.mit.edu/research/evaporative-cooling-vegetable-preservation. Cited in ZBX-PRB-001.
- MIT D-Lab. "Room-Sized Cooling Chambers." https://d-lab.mit.edu/research/evaporative-cooling-vegetable-preservation/room-sized-cooling-chambers. Accessed October 24, 2026.

- MIT News. "MIT D-Lab and CITE Release Evaluation of Low-Cost Evaporative Cooling Devices in Mali." 2018. https://news.mit.edu/2018/mit-d-lab-cite-evaluation-low-cost-evaporative-cooling-devices-mali-0620.
- Munters. "CELdek." https://piec.com/celdek/. Cited in ZBX-PRS-001.
- Review of evaporative pad operation. *Renewable and Sustainable Energy Reviews*. https://www.sciencedirect.com/science/article/pii/S1364032121009072. Cited in ZBX-PRB-001.
- Stull, Roland. "Wet-Bulb Temperature from Relative Humidity and Air Temperature." *Journal of Applied Meteorology and Climatology* 50 (2011): 2267 to 2269.
- TechXLab. "Cold Rooms by ColdHubs." https://techxlab.org/solutions/cold-rooms-by-coldhubs/. Accessed October 24, 2026.
- *TIME*. "Best Inventions of 2001: The Pot-in-Pot Cooler." https://content.time.com/time/specials/packages/article/0,28804,1936165_1936254_1936632,00.html. Cited in ZBX-PRB-001.
- UC Davis Postharvest Center. "Tomato." https://postharvest.ucdavis.edu/produce-facts-sheets/tomato. Cited in ZBX-PRB-001.
- Verploegen, Eric, Ousmane Sanogo and Takemore Chagomoka. *Evaluation of Low-Cost Evaporative Cooling Technologies for Improved Vegetable Storage in Mali*. MIT D-Lab, GHTC paper. https://d-lab.mit.edu/sites/default/files/inline-files/GHTC%20-%20Evaluation%20of%20Low-Cost%20Evaporative%20Cooling%20Technologies%20for%20Improved%20Vegetable%20Storage%20in%20Mali.pdf.
- Verploegen, Eric, Ousmane Sanogo and Takemore Chagomoka. *Evaporative Cooling Technologies for Improved Vegetable Storage in Mali, Evaluation Report*. MIT D-Lab, June 2018. https://d-lab.mit.edu/sites/default/files/inline-files/Evaporative%20Cooling%20Technologies%20for%20Improved%20Vegetable%20Storage%20in%20Mali%20-%20Evaluation%20Report_0.pdf. Accessed October 24, 2026.
- Wayua, Francis O., Michael W. Okoth and John Wangoh. "Design and Performance Assessment of a Low Cost Evaporative Cooler for Storage of Camel Milk in Arid Pastoral Areas of Kenya." De Gruyter, 2012. https://www.degruyterbrill.com/document/doi/10.1515/1556-3758.2323/pdf. Accessed October 24, 2026.
- World Bank. "Is Post-Harvest Loss Significant in Sub-Saharan Africa?" *Africa Myths and Facts*. https://www.worldbank.org/en/programs/africa-myths-and-facts/publication/is-post-harvest-loss-significant-in-sub-saharan-africa.
- World Health Organization. *Legionella and the Prevention of Legionellosis*. Geneva: WHO, 2007. https://www.who.int/publications/i/item/9241562978. Accessed October 24, 2026.